

Assignment 1

Here are some basic problems on single-qubit gates to get you started. Make sure to work them out by hand!

1. Single qubit gates are expressed as 2×2 unitaries. Ignoring a phase factor $e^{i\phi}$, every single qubit unitary can also be expressed as a rotation on the Bloch Sphere around an axis $\hat{n}$ by an angle θ , denoted $R_{\hat{n}}(\theta)$. Show with steps the axis and angle for each of the gates: X, Y, Z, H, S, T
2. Prove the following identities, which are useful for simplifying circuits:
 $HXH = Z, HYH = -Y, HZH = X$
3. Construct a quantum circuit to add two-bit numbers x and y modulo 4. That is, the circuit takes in two bits of x and two bits of y , applies a sequence of some gates, and outputs the 4-bit sum of x and y
4. Decompose the following unitary into a product of two-level unitaries:

$$\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$